

HS300 drive — maintenance manual

A 22 kW IGBT inverter driving a submersible pump from a PV array, from a battery bus, or from a weak grid, depending on the build. Sensorless field-oriented control with its own auto-tune; no encoder, nothing down the well but the motor.

What you need in front of you: a laptop with this folder, a USB-RS485 adapter (or the ST-Link's USB serial), and a multimeter.

Tell the scripts where the drive is — once

```
PORT.bat
```

Double-click it. It lists the serial ports it can see, asks which one the drive is on and which cable you are using, and makes the setting for you. Run it again to check or change it.

FLASH.bat, CHECK_SENSOR.bat, ZERO_ADC.bat and DASHBOARD.bat then all find the drive on their own and each says which setting it obeyed. If you start one before setting a port, it asks you there and then - you are never left stuck. A port typed on the command line always wins.

The setting is an empty file whose **name** is the whole thing:

```
tty4.485      COM4 over the USB-RS485 adapter
tty5.ttl      COM5 over the ST-Link USB serial / direct TTL
```

Keep one, not two - PORT.bat replaces the old one when you change it, because whichever is found first would otherwise win and the other is a trap. Put it here or one level up; both are searched.

It is tty, not com. Windows reserves COM1..COM9 as device names, over the whole name before the first dot, so File Explorer refuses com5.ttl as an invalid name. Some other tools create it anyway, which is worse rather than better - the file exists but Explorer may then refuse to rename or delete it. port5.ttl and ccom5.ttl also work, as does .rs485 for .485.

A setting whose name cannot be read as a port is **reported**, not ignored, and you are offered a replacement. One that cannot be understood must not look the same as none at all.

The dashboard

```
DASHBOARD.bat
```

Starts the agent and opens <http://localhost:8080>. Live telemetry, settings you can edit, and the commands. This is the easiest way to see what the drive is doing.

It holds the serial port while it runs. Nothing else can talk to the drive at the same time — close that window before FLASH.bat, CHECK_SENSOR.bat or ZERO_ADC.bat.

In order, on a drive you have not met before

1. CHECK_SENSOR.bat — the DC current sensor's range. Do this first, every time, on any board you have not personally checked. Reason in the next section.
2. ZERO_ADC.bat — only with the **DC bus at zero volts**. It measures the offsets as they sit, so whatever is present becomes "zero" for ever.
3. FLASH.bat <BOARD> <COM> — the proven image for that board.
4. DASHBOARD.bat — check the plate, then run a tune, then a test run.

Why CHECK_SENSOR comes first

New boards leave the factory with the DC current **offset pot at 2.5 V** and it must be turned down to about **0.5 V**. On a 3.3 V ADC a 2.5 V rest point leaves only about **27 A** of measurable current, and the sensor then simply

reports a flat maximum — no fault, no warning.

Because the MPPT verdict is $V_{dc} \times I_{dc}$, once the current rails the tracker sees power **fall** whenever it loads harder, so it parks on that false ceiling and reports it as the array's limit. One board ran that way for a whole morning with roughly a tenth of the array unused and nothing complaining.

CHECK_SENSOR.bat catches it in two minutes. If it says FAIL, turn the pot down until the rest point is near the bottom of the range, then ZERO_ADC.bat.

Settings worth knowing

Edit from the dashboard, or by name with the tools. **Over the wire they are in real units** — the panel's F-number and its $\times 10$ scaling are a different thing.

setting	what it is	watch out
PLATE_KW, PLATE_V_LL, PLATE_F_HZ, PLATE_I_RMS, PLATE_PP, PLATE_RPM	the motor nameplate	everything derives from this: the tune's current levels, the stall gate, i_open. A wrong plate is the most common cause of a tune that will not pass
F01 OVER_CURRENT	the trip	must be above what the pre-start itself draws, or the check trips
F03 OVER_VOLTAGE	bus OV trip	check it is reachable — on one board it sat above what the sensor could report
F05 OVER_TEMP	heatsink trip, 90 °C	59 °C at full load on the 37K is normal
F08 FMIN / F09 FMAX	frequency limits	FMAX is the plate frequency. Going past it over-speeds the pump
F10 ACC_FACTOR	ramp rate	
MODE	0 manual, 1 MPPT, 2 DC bus	change it on the fly. Stopping and restarting to change mode makes the tracker re-enter from the floor and the drive dives to minimum frequency
AUTOSTART_EN	starts on its own when the bus comes up	leave 0 unless you mean it. On a battery bus this once drained the pack flat overnight

Settings do not persist unless you save them

A write changes the live value only. **Without SAVE_SETTINGS it is lost at the next reset**, and these boards reset often. The dashboard's save button sends it; by hand it is CMD_SAVE_SETTINGS.

To **prove** a value is stored: save, then CMD_RELOAD_EEPROM, then read it back. A plain read-back straight after a write tells you nothing.

Normal readings, 22 kW well machine at full sun

f	48-50 Hz	Vdc	650-680 V
Idc	30-34 A	Pdc	20-22 kW
psi_r	0.985-0.99	iq	45-53 A peak
temp	55-60 C	BUS_SHED	1.000

BUS_SHED below 1.0 means the bus sagged and the drive softened the load. Brief dips on a cloud edge are the design working. Sitting at 0.2 is not.

Flashing

FLASH.bat <BOARD> <COM> uses the **STABLE** image for that board. It must print **both** of these:

```
verified and marked bootable
NEW app is up
```

If you do not see both, the drive did not take it.

- **An OTA that stalls part way needs SWD recovery, not another attempt.** Stop and ask the engineer.
- A **virgin board cannot be flashed over serial at all** — there is no bootloader yet. Use `full_<BOARD>_<CRC>.hex` with an ST-Link and **mass erase the chip first**, or stale metadata will make the bootloader refuse the app.
- A board with **no STABLE** has never been proven. Ask; do not flash LATEST.

Keep a record

`drive_backup.py` saves the EEPROM settings and the firmware CRC before you change anything. Do it before a tune or a flash — it is the only way to get a drive back to how it was.